

Topic B16

Inheritance

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B17.1	Inheritance of genetic information
B17.2	Chromosomes, genes and proteins
B17.3	Mitosis
B17.4	Meiosis
B17.5	Monohybrid inheritance
B17.6	Sex determination

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Genetic information

A chromosome is a DNA molecule carrying genes. A gene is a length of DNA coding for a protein; an allele is an alternative form of a gene. Base sequence determines amino-acid sequence and protein shape. Most body cells are diploid with homologous chromosome pairs.

Examiner insight

Do not use gene and allele as synonyms.

2. Mitosis

Before mitosis DNA replicates. Nuclear division produces genetically identical nuclei with the same chromosome number, followed by cell division. Mitosis enables growth, replacement, repair and asexual reproduction. Uncontrolled division can form tumours.

Examiner insight

Mitosis maintains chromosome number; it does not generate gametes.

3. Meiosis and variation

Meiosis occurs in reproductive organs and produces four genetically different haploid cells. Chromosome number is halved; independent assortment and crossing over create variation. Fertilisation restores diploid number and combines alleles from two parents.

Examiner insight

Variation is produced by meiosis and random fertilisation, not by mitosis.

4. Monohybrid inheritance

A genotype is the allele combination; phenotype is the expressed feature. A dominant allele is expressed in a heterozygote; a recessive allele requires two copies. Genetic diagrams show parental genotypes, gametes, offspring combinations and predicted ratios.

Examiner insight

Probabilities predict many offspring; they do not guarantee the outcome of a small family.

Worked application

Situation: Two heterozygous parents carry a recessive condition.

Prediction: Each child has a 25% probability of being affected.

Reason: The cross gives aa, aa, aa and aa in a 1:2:1 genotype ratio.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B16

Inheritance

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines gene? [1]
 - A a length of DNA that codes for a protein
 - B an alternative form of a gene
 - C the alleles an organism possesses
 - D observable features resulting from genotype and environment
2. Which statement best defines allele? [1]
 - A the alleles an organism possesses
 - B an alternative form of a gene
 - C observable features resulting from genotype and environment
 - D having two different alleles of a gene
3. Which statement best defines genotype? [1]
 - A observable features resulting from genotype and environment
 - B having two different alleles of a gene
 - C the alleles an organism possesses
 - D a length of DNA that codes for a protein
4. Which statement best defines phenotype? [1]
 - A having two different alleles of a gene
 - B a length of DNA that codes for a protein
 - C an alternative form of a gene
 - D observable features resulting from genotype and environment
5. Which statement best defines heterozygous? [1]
 - A having two different alleles of a gene
 - B a length of DNA that codes for a protein
 - C an alternative form of a gene
 - D the alleles an organism possesses
6. Which statement about genetic information is correct? [1]
 - A An entire nucleus.
 - B A chromosome is a DNA molecule carrying genes.
 - C A pair of chromosomes.
 - D The visible expression only.
7. Which statement about mitosis is correct? [1]
 - A A pair of chromosomes.
 - B The visible expression only.
 - C Before mitosis DNA replicates.
 - D The allele symbols only.

8. Which statement about meiosis and variation is correct? [1]
- A The visible expression only.
 - B The allele symbols only.
 - C Having two identical recessive alleles.
 - D Meiosis occurs in reproductive organs and produces four genetically different haploid cells.
9. Which statement about monohybrid inheritance is correct? [1]
- A A genotype is the allele combination; phenotype is the expressed feature.
 - B The allele symbols only.
 - C Having two identical recessive alleles.
 - D An entire nucleus.
10. Which prediction is most appropriate when two heterozygous parents carry a recessive condition? [1]
- A An entire nucleus.
 - B The visible expression only.
 - C Each child has a 25% probability of being affected.
 - D Having two identical recessive alleles.

Section B · Structured questions

11 Biologists use precise vocabulary when studying inheritance.

(a) Define gene.

[2]

(b) Define allele.

[2]

(c) Distinguish genotype from the incorrect idea: “the visible expression only”.

[2]

(d) State two important features of genetic information.

[2]

[Total: 8]

12 Two heterozygous parents carry a recessive condition.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records percentage observed under three conditions.

condition	percentage observed
10 offspring	20
50 offspring	24
200 offspring	25

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to model inheritance by crossing heterozygous tall pea plants.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of inheritance to an unfamiliar context.

(a) Define phenotype. [2]

(b) Explain the biological importance of meiosis and variation. [3]

(c) Correct this statement: “having two identical recessive alleles”.

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. A length of dna that codes for a protein.
- 2 B. An alternative form of a gene.
- 3 C. The alleles an organism possesses.
- 4 D. Observable features resulting from genotype and environment.
- 5 A. Having two different alleles of a gene.
- 6 B. Do not use gene and allele as synonyms.
- 7 C. Mitosis maintains chromosome number; it does not generate gametes.
- 8 D. Variation is produced by meiosis and random fertilisation, not by mitosis.
- 9 A. Probabilities predict many offspring; they do not guarantee the outcome of a small family.
- 10 C. The cross gives aa, aa, aa and aa in a 1:2:1 genotype ratio.

Solutions · Structured questions

11. (a) a length of DNA that codes for a protein [2]. (b) an alternative form of a gene [2]. (c) the alleles an organism possesses; not the visible expression only [2]. (d) Credit any two accurate features from the guide: A chromosome is a DNA molecule carrying genes. A gene is a length of DNA coding for a protein; an allele is an alternative form of a gene. Base sequence determines amino-acid sequence and protein shape. Most body cells are diploid with homologous chromosome pairs. [2].

12. (a) Each child has a 25% probability of being affected [2]. (b) The cross gives aa, aa, aa and aa in a 1:2:1 genotype ratio [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $25 - 20 = 5$ [2]. (b) The values change from 20 to 25, with the highest value 25. [2]. (c) Larger samples tend to approach the predicted 25% from a heterozygous monohybrid cross [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Random gamete combination [1]. (b) Phenotype ratio [1]. (c) Same parental genotype and scoring criteria [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Increase offspring number and repeat crosses [1].

15. (a) observable features resulting from genotype and environment [2]. (b) Award linked points from: Meiosis occurs in reproductive organs and produces four genetically different haploid cells. Chromosome number is halved; independent assortment and crossing over create variation. Fertilisation restores diploid number and combines alleles from two parents. [3]. (c) Correct statement: heterozygous means having two different alleles of a gene [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.